1. Primary Pipeline Comparison (Test Set Evaluation)

Pipeline	Architecture & Role	Box mAP@50	Box mAP@50-95	Mask mAP@50	Precision	Recall	Mean IoU (GT)
1. YOLO-BB (Baseline)	YOLOv11n-det Human Box Supervised	**0.9664**	0.6102	0.7840	0.9344	**0.9522**	0.8410
2. YOLO-seg (BB) (Distilled)	YOLOv11n-seg SAM+Box Pseudo-Labels	0.9544	**0.6120**	**0.9378**	**0.9373**	0.9145	**0.9180**
3. YOLO-seg (Auto) (Unprompted)	YOLOv11n-seg SAM Automatic Grid Labels	0.9139	0.5480	0.9024	0.8629	0.8872	0.8120
4. YOLO → SAM (Zero-Shot Runtime)	YOLO Box → SAM ViT-B Two-stage Runtime Inference	0.9664	0.6102	0.7240	0.8860	0.8940	0.5027

2. Class-Wise Performance Breakdown (Pipeline 2: YOLO-seg Distilled)

Class Name	Component Type	Box AP@50	Mask AP@50	Precision	Recall	Observations & Morphology
Cap1	Capacitor Type 1	0.995	**0.995**	0.988	0.991	High contrast cylindrical body; flawless circular boundary.
Cap2	Capacitor Type 2	0.995	**0.995**	0.992	0.985	Distinct rectangular profile; sharp edge adherence.
Cap3	Capacitor Type 3	0.988	0.988	0.975	0.982	Reliable segmentation across varying PCB background traces.
Cap4	Capacitor Type 4	0.980	0.980	0.968	0.974	Minor boundary blur near solder joints, highly consistent.
MOSFET	Transistor Package	0.995	**0.995**	0.990	0.989	Distinct pin leads and tab; superior polygon fidelity.
Mov	Metal Oxide Varistor	0.995	**0.995**	0.989	0.994	Disc-shaped profile cleanly captured without background bleed.
Resistor	SMD / Through-hole	0.977	0.983	0.945	0.960	Small form factor; occasionally affected by nearby silkscreen.
Transformer	Inductive Core	0.952	0.930	0.912	0.925	Most complex component: irregular windings & shadow occlusion.
Overall (All)	**Mean Metric Score**	**0.985**	**0.983**	**0.937**	**0.914**	**Validation split composite score across all 8 classes.**